

Agnes Namyalo

PhD Researcher, Computing (Data Science) | Boise State University
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RESEARCH INTERESTS

My research sits at the intersection of computational text analysis, natural language processing, and policy communication. I develop automated pipelines using large language models to extract theoretically grounded narrative elements from policy texts, and apply statistical models to test whether media coverage of Endangered Species Act events predicts congressional attention. My work contributes reusable research software, validated prompt engineering methodology, and new empirical evidence on media-congress agenda-setting in environmental policy.

EDUCATION

PhD in Computing (Data Science) Aug 2024 – Present

Boise State University, Boise, Idaho, USA

- **Dissertation: Computational Analysis of Endangered Species Act Narratives** — Media, Congress, and Policy Outcomes
- Developing an LLM-based pipeline to extract Narrative Policy Framework elements from text documents
- Applying statistical models and machine learning methods to test media agenda-setting hypotheses
- **Teaching Assistant: CS 121** — Introduction to Computer Science

Professional Certificate — Data Science and Machine Learning Mar 2023 – Jun 2023

Massachusetts Institute of Technology / Great Learning

- Python, regression, classification, hypothesis testing, deep learning, recommendation systems

MSc in Mathematics of New Material Design and Nanotechnology Aug 2019 – Oct 2020

Gdansk University of Technology, Poland

- Computer modeling, classical and quantum simulations, stochastic methods

MSc in Mathematical Engineering (Scientific Computing) Aug 2018 – Oct 2019

University of L'Aquila, Italy

- Control systems, dynamical systems, stochastic modeling, numerical methods, data analysis

BSc in Mathematics and Computer Science Aug 2012 – Oct 2015

Busitema University, Uganda

- Statistics, probability, algorithms, database systems, simulation and modeling
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RESEARCH EXPERIENCE

PhD Researcher — Boise State University Aug 2024 – Present

Dissertation: Computational Analysis of Endangered Species Act Narratives

- **LLM extraction pipeline (Python+ Claude API)** — five-layer prompt system extracting NPF narrative elements (event type, setting, characters, plot, moral) from ESA news and congressional documents; includes modular prompt builders, batch API calls, JSON parsing, and reproducible CSV outputs
- **Statistical agenda-setting analysis** — fitted OLS, Poisson, Quasi-Poisson, Negative Binomial, Hurdle regression models and machine learning related models to test whether ESA coverage predicts congressional ESA activity
- **Prompt engineering methodology** — developed and validated a general-purpose NPF extraction prompt with consistency rules.

Data Scientist — Dfcu Bank, Kampala, Uganda Jun 2022 – Aug 2024

- Built Python + SSIS pipeline extracting and integrating core banking data to data warehouse, improving forecasting reliability across retail and corporate divisions
- **Automated bank reconciliation process (bank-to-wallet transactions) using Power Automate and Python** — reduced processing time from 2 hours to 3 minutes
- Developed customer churn prediction model using logistic regression and Random Forest (pandas, NumPy, scikit-learn) enabling targeted retention interventions

Data Scientist — Centre for Computational Biology, Uganda Christian University Jun 2021 – Oct 2021

- Built COVID-19 time series forecast model for Uganda using Facebook Prophet library and regression techniques
- Developed data extraction scripts and statistical visualizations (pandas, matplotlib) for weekly, monthly, and regional case analysis

Business Intelligence Analyst — Equity Bank, Kampala, Uganda Jan 2022 – May 2022

- Integrated BI assets into CRM workflows; collected and maintained data from multiple sources using database technologies

TEACHING EXPERIENCE

Teaching Assistant — CS 121: Introduction to Computer Science Aug 2024 – Present

Boise State University

- Supported undergraduate students in building foundational programming and computational thinking skills in Python
- Conducted office hours; contributed to an accessible and engaging learning environment
- Helped students from diverse backgrounds develop confidence with computational problem-solving

TECHNICAL SKILLS

Programming Languages

R (tidyverse, dplyr, ggplot2, vars) • Python (pandas, numpy, matplotlib, scikit-learn, sentence-transformers, openai) • SQL • Bash

Research Software and Tools

Git/GitHub • RStudio • VS Code • Jupyter Notebooks • Power Automate • SSIS • LaTeX (Overleaf, ACM format)

Statistical and Machine Learning Methods

Count regression (Poisson, Quasi-Poisson, Negative Binomial, Hurdle) • Structural Topic Modeling • Large Language Model prompt engineering • Logistic regression • Random Forest • Time series forecasting • Granger causality testing

Languages

English (fluent) • Luganda (native) • Italian (intermediate)